

PIP-CN: Prediction-Integrated Planning for Crowd Navigation

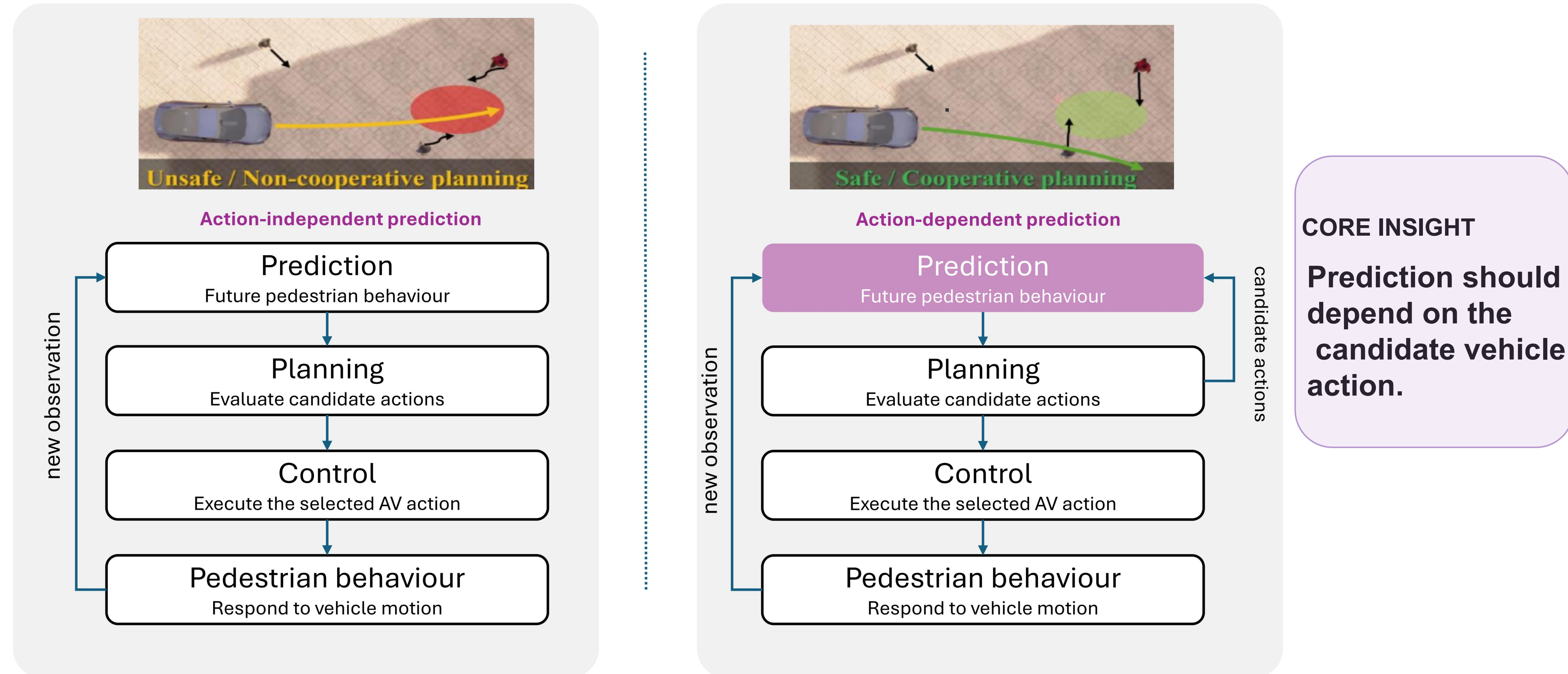
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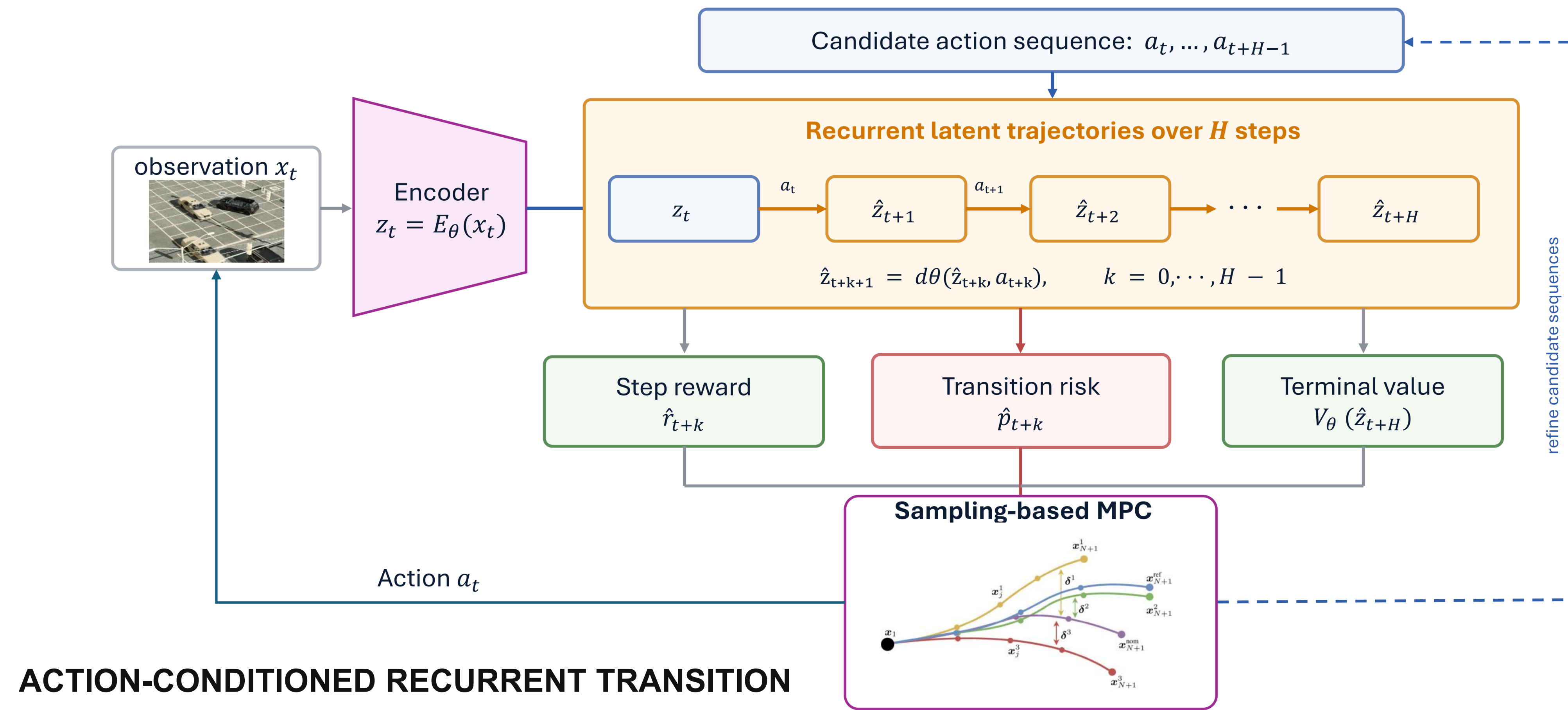
² University of Nottingham



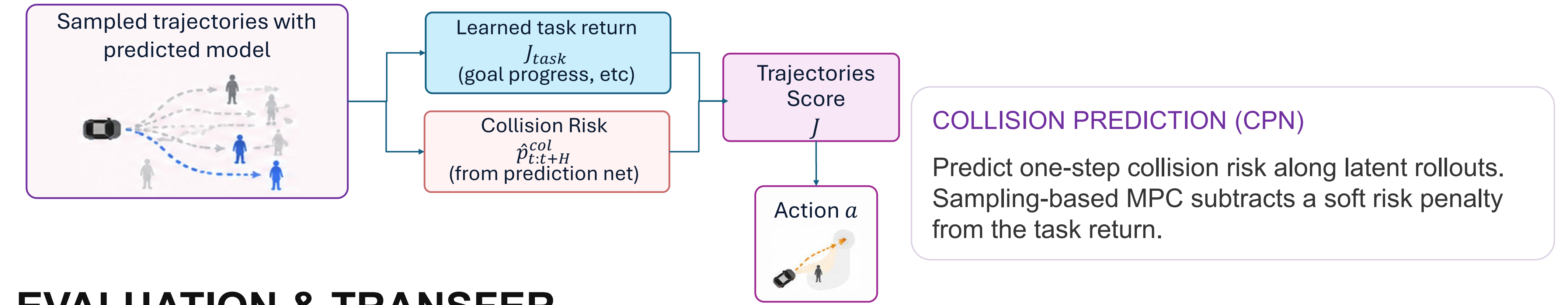
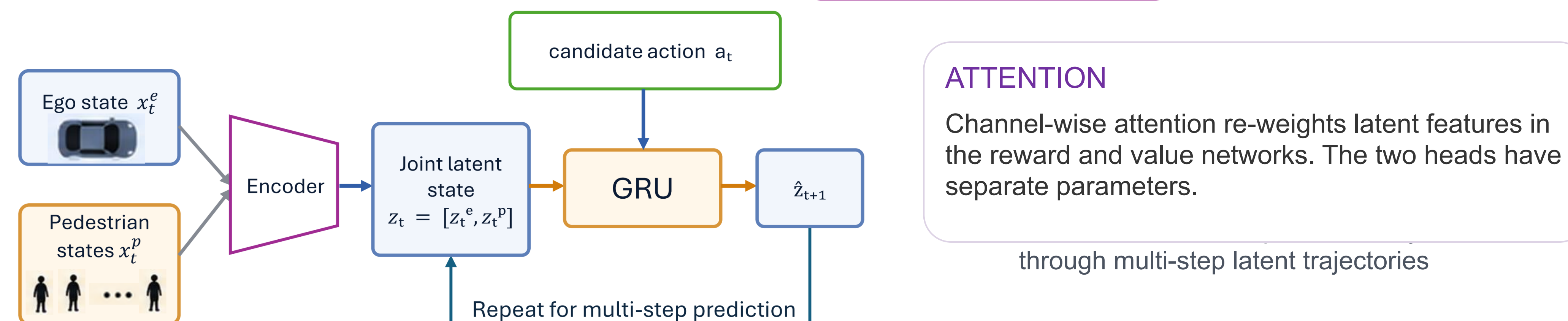
WHY ACTION-DEPENDENT PREDICTION?



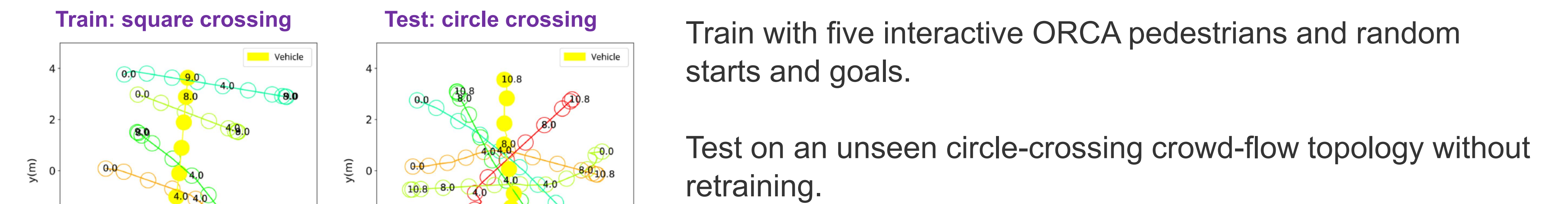
PIP-CN ARCHITECTURE



ACTION-CONDITIONED RECURRENT TRANSITION



EVALUATION & TRANSFER



QUANTITATIVE RESULTS

Method	Square-crossing scenario					Circle-crossing scenario (OOD)				
	SR[%]↑	CR[%]↓	ET[s]↓	FD[%]↓	MD[m]↑	SR[%]↑	CR[%]↓	ET[s]↓	FD[%]↓	MD[m]↑
TD-MPC	97.2	2.8	8.59 ± 0.68	3.1	0.17	95.1	4.9	10.49 ± 1.84	3.7	0.18
Ours (no CPN, no att)	99.9	0.0	8.48 ± 0.54	19.4	0.16	99.8	0.2	8.47 ± 0.28	6.8	0.14
Ours (no CPN, with att)	99.8	0.2	8.21 ± 0.12	0.0	-	99.9	0.0	8.21 ± 0.07	0.0	-
Ours	99.9	0.0	9.21 ± 0.95	2.7	0.18	99.9	0.0	8.63 ± 0.45	3.8	0.19

SR: success rate; CR: collision rate; ET: expected time; FD: proportion of danger steps; MD: mean minimum distance.

ORCA reports zero collisions by design. "-" retained from source.

CARLA: QUALITATIVE COMPARISON

